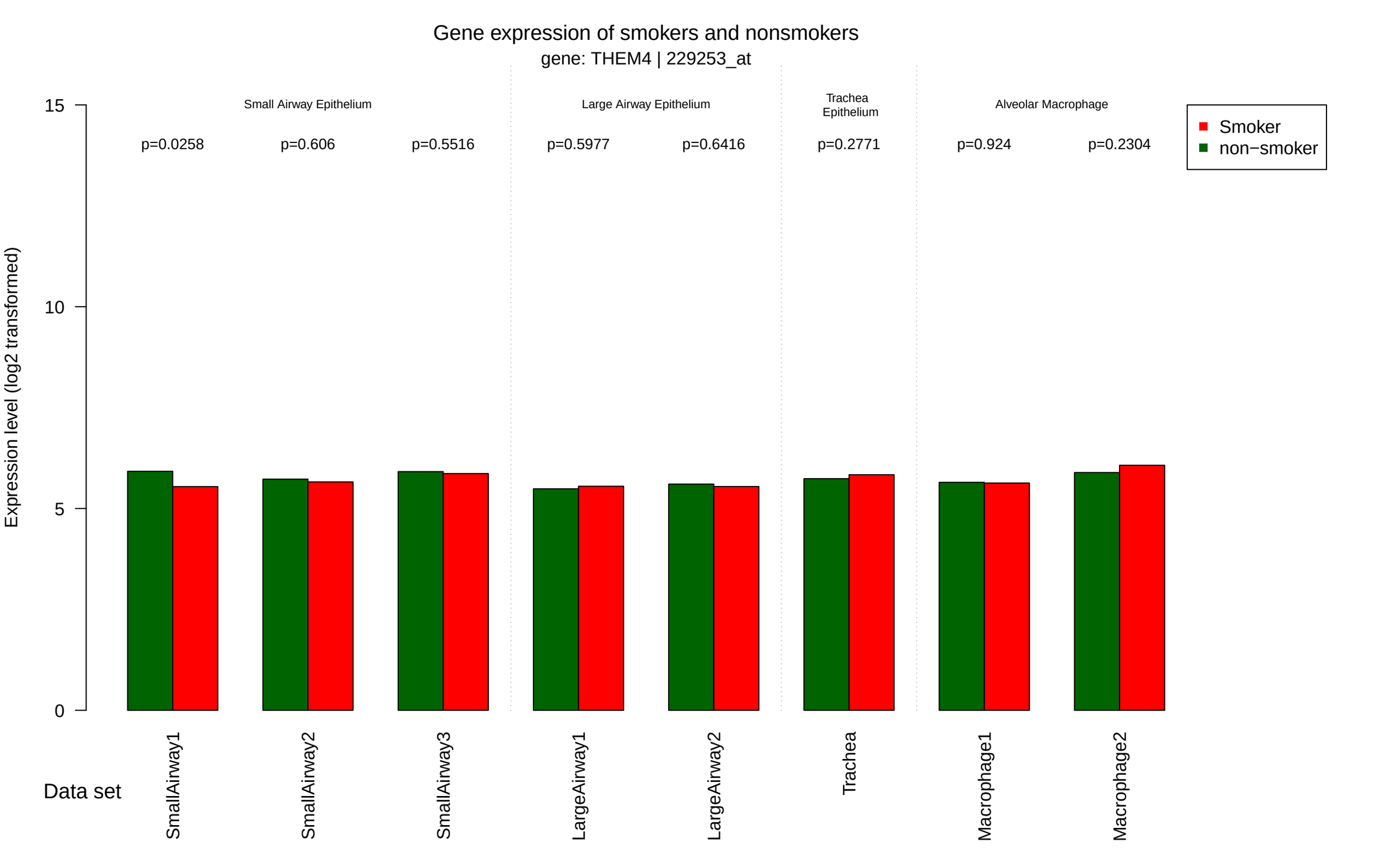
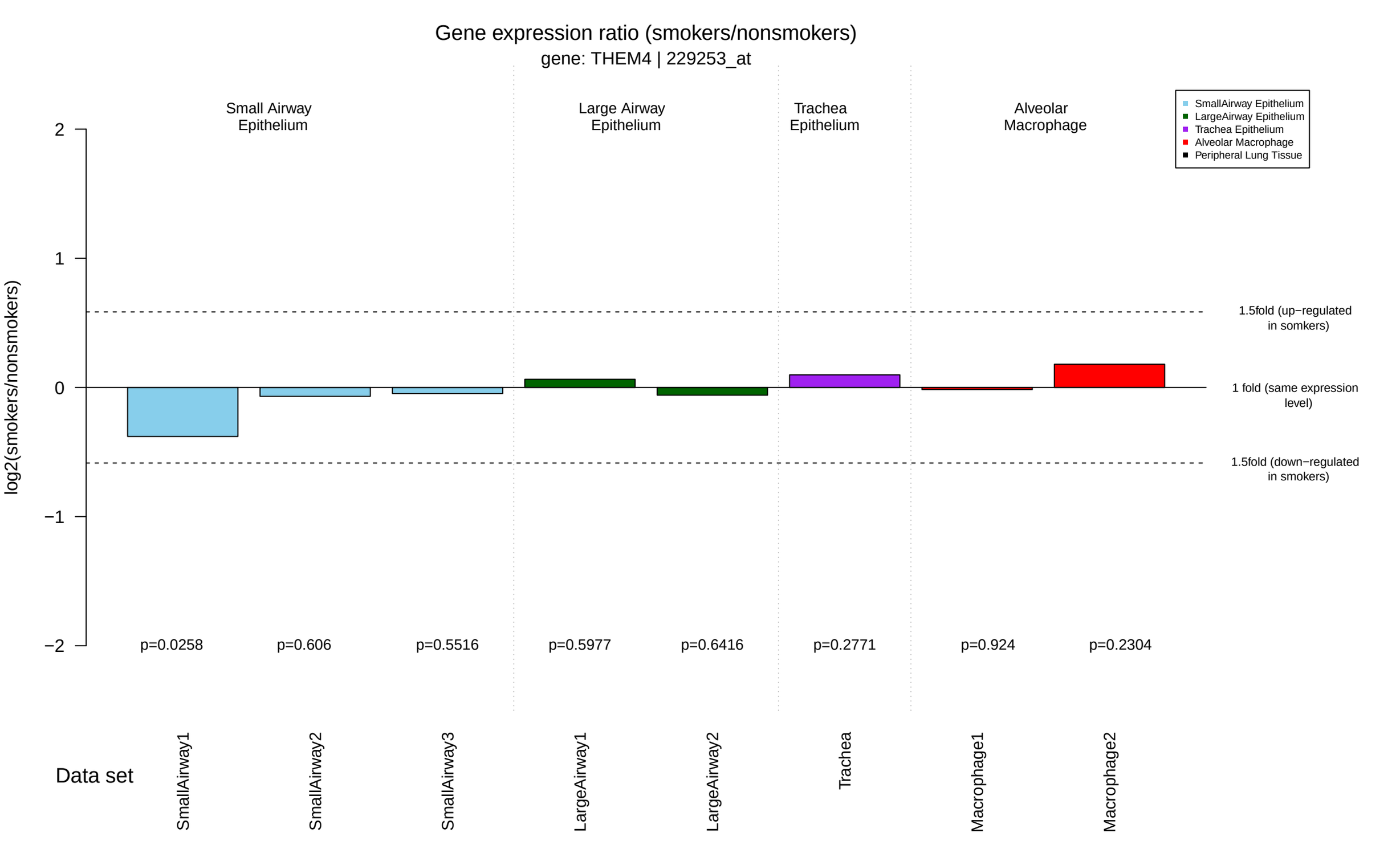
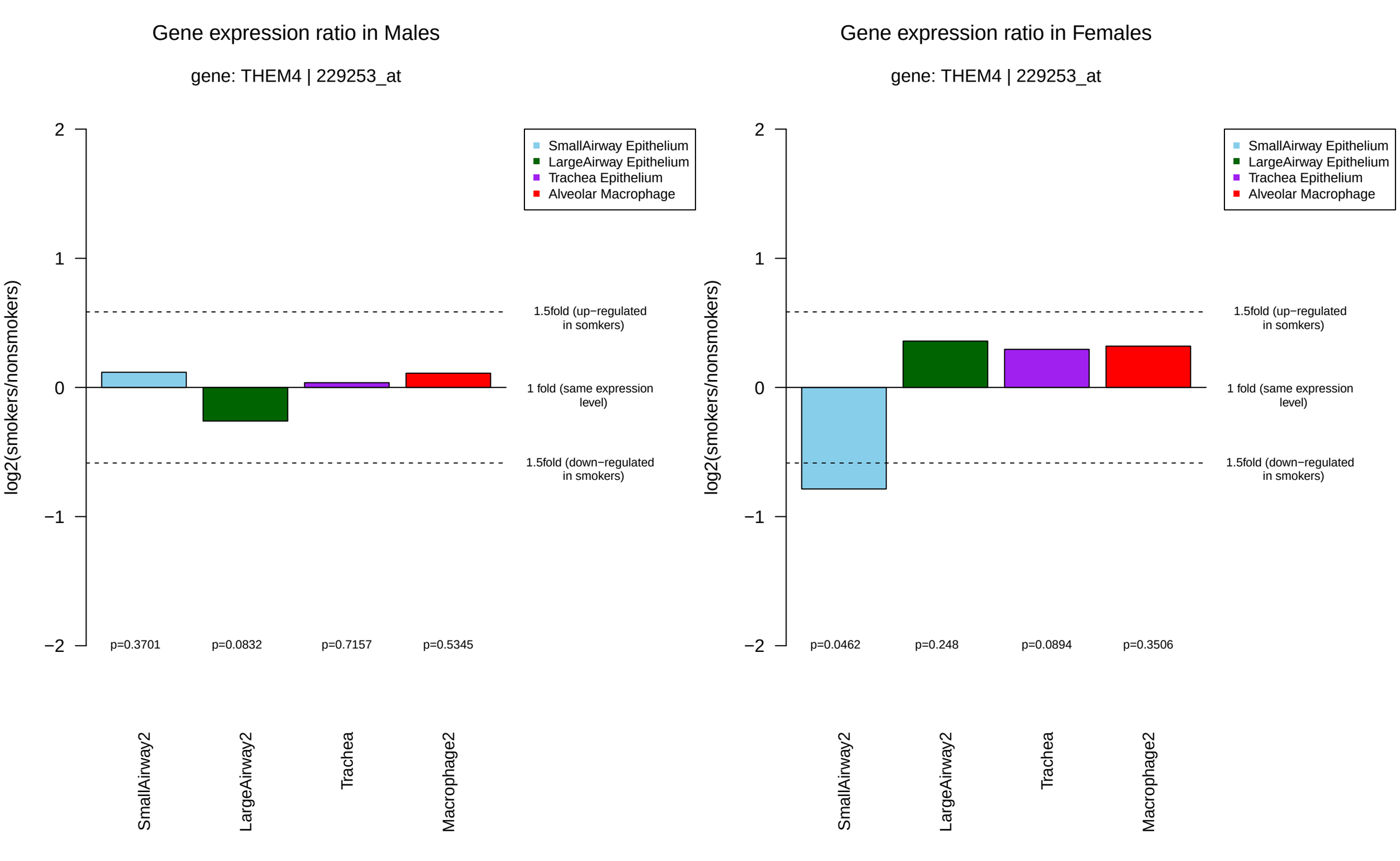
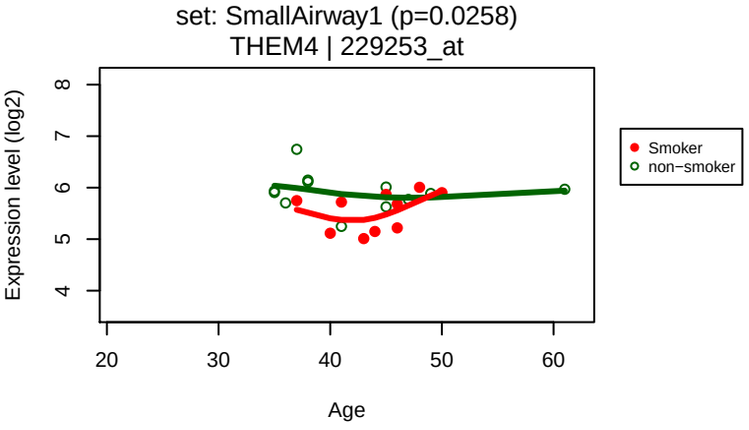
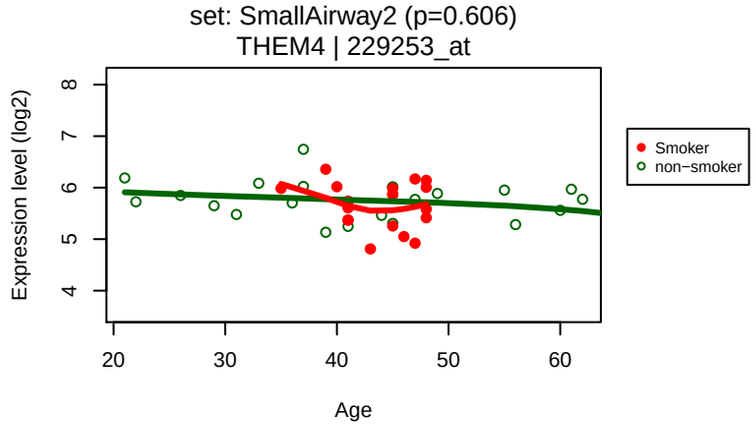
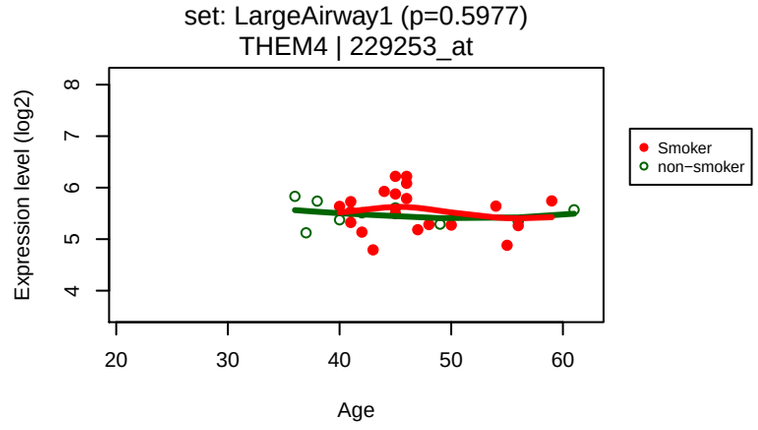

Gene expression of smokers and nonsmokers
gene: THEM4 | 229253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0258
p=0.606
p=0.5516
p=0.5977
p=0.6416
p=0.2771
p=0.924
p=0.2304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

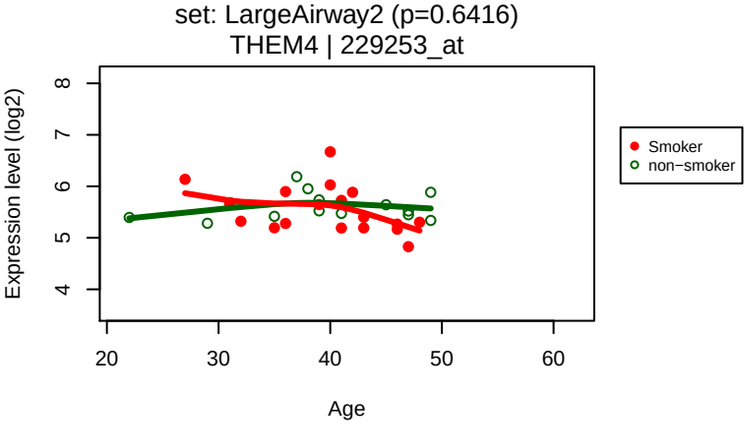

Gene expression ratio (smokers/nonsmokers)
gene: THEM4 | 229253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0258
p=0.606
p=0.5516
p=0.5977
p=0.6416
p=0.2771
p=0.924
p=0.2304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

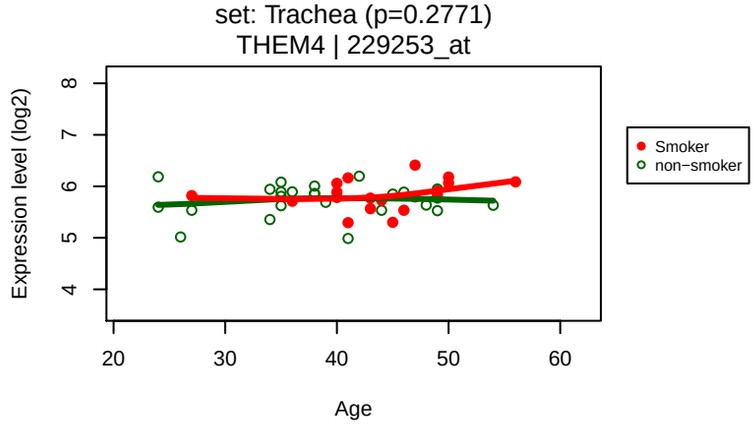

Gene expression ratio in Males
gene: THEM4 | 229253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3701
p=0.0832
p=0.7157
p=0.5345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: THEM4 | 229253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0462
p=0.248
p=0.0894
p=0.3506
SmallAirway2
LargeAirway2
Trachea
Macrophage2

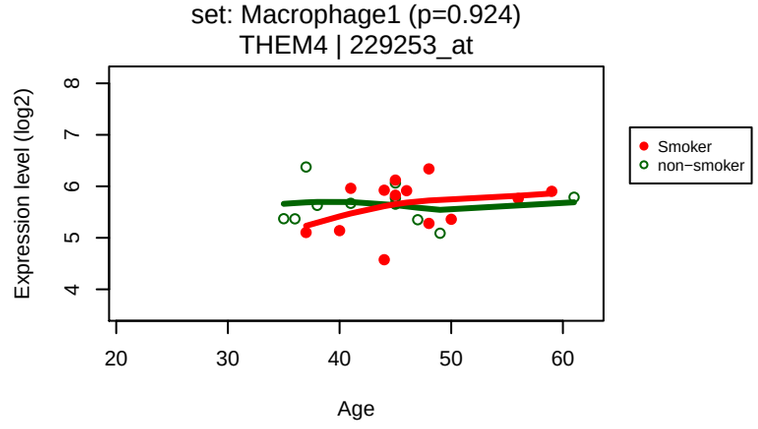

set: SmallAirway1 (p=0.0258)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker

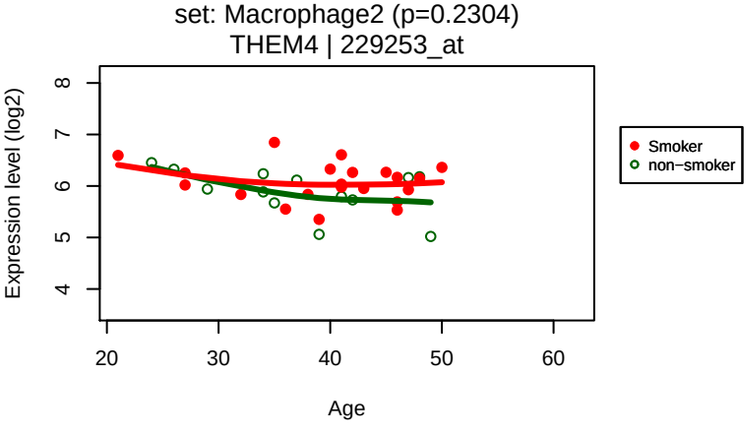

set: SmallAirway2 (p=0.606)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker

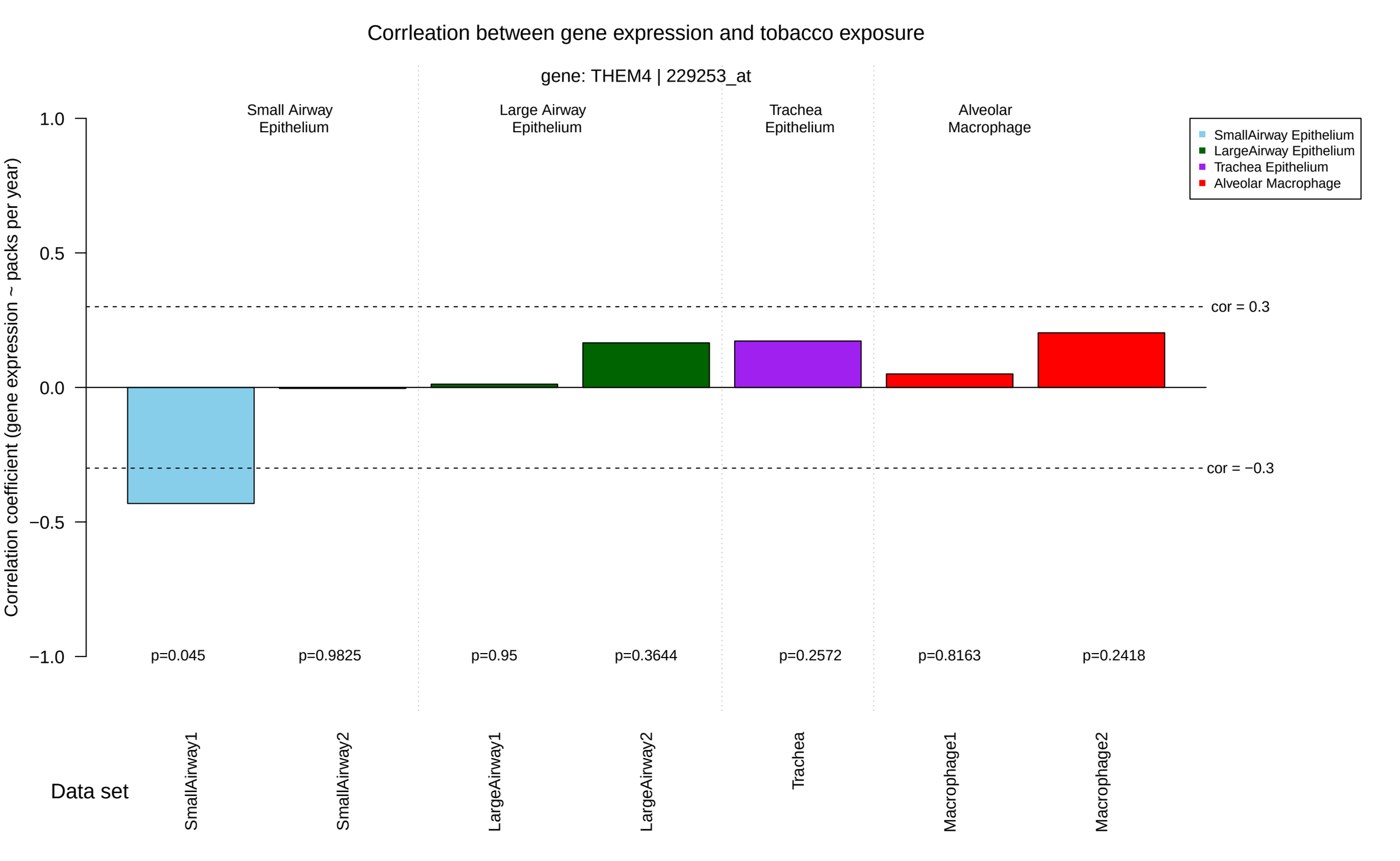

set: LargeAirway1 (p=0.5977)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6416)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2771)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.924)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2304)
THEM4 | 229253_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: THEM4 | 229253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.045
p=0.9825
p=0.95
p=0.3644
p=0.2572
p=0.8163
p=0.2418
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2